

ECE385

DIGITAL SYSTEMS LABORATORY

Introduction to DDR3 on the Urbana Board

Abstract and Goals

The goal of this document is to assist students in creating a DDR3 based design using the Vivado MIG (Memory Interface Generator). This allows access to the 64M x 16 (1 Gigabit) DDR3 external ram chip which is located on the Urbana board. As the MIG creates a memory controller with an AXI4 slave port, this document uses the MicroBlaze processor to test the functionality of the external memory. If your design requires direct hardware access to the DDR3 chip, you will need to make the AXI port external and create an AXI bus master.

Setting up the Memory Interface Generator:

Create a new project and create a new block design. Follow the steps from the previous MicroBlaze related labs and create a basic (empty) block design. Add the MicroBlaze processor to your blank block design as before, but **do not** select the “Run Block Automation”. Instead, we will create the memory controller first. Add the “Memory Interface Generator (MIG 7 Series)”, and double click on the block.

Select next on the first customization screen for the MIG and then select “Create Design” and click next again:

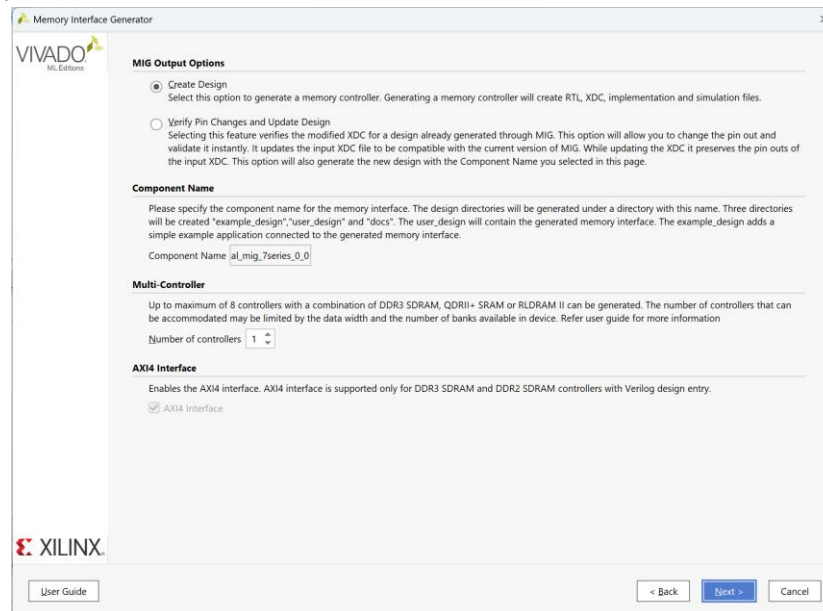


Figure 1 - MIG Intro

Click Next on the following screen (you do not need to check any of the options, as this is only used if you need to target the smaller version of the FPGA – Spartan 7 with 25K LUTs as well).

Since we are only targeting a single part (Spartan 7 – 50K) we don't need to maintain cross compatibility. On the following screen, select DDR3, and click Next. Finally, the next screen is the first where you must change a significant number of settings as follows:

Memory Interface Generator

VIVADO ML Editions

Options for Controller 0 - DDR3 SDRAM

Clock Period: Choose the clock period for the desired frequency. The allowed period range(3000 - 3300) is a function of the selected FPGA part and FPGA speed grade. Refer to the User Guide for more information. 3,000 ps 333.33 MHz

PHY to Controller Clock Ratio: Select the PHY to Memory Controller clock ratio. The PHY operates at the Memory Clock Period chosen above. The controller operates at either 1/4 or 1/2 of the PHY rate. The selected Memory Clock Period will limit the choices. 4:1

Memory Type: Select the memory type. Type(s) marked with a warning symbol are not compatible with the frequency selection above. Components

Memory Part: Select the memory part. Part(s) marked with a warning symbol are not compatible with the frequency selection above. Find an equivalent part or create a part using the "Create Custom Part" button if the part needed is not listed here. The "Create Custom Part" feature is not supported for RDRAM II. MT41K64M16XX-125 Create Custom Part

Memory Voltage: Select the Voltage of the Memory part selected. 1.35V

Data Width: Select the Data Width. Parts marked with a warning symbol are not compatible with the frequency and memory part selected above. 16

ECC: MIG supports ECC for 72 bit data width configuration. To be able to select ECC, select a data width that has ECC supported. Disabled

Data Mask: Enable or disable the generation of Data Mask (DM) pins using this check box. This option can be selectable only if the memory part selected has DM pins. Uncheck this box to not use data masks and save FPGA I/Os that are used for DM signals. ECC designs (DDR3 SDRAM, DDR2 SDRAM) will not use Data Mask. ☒

Number of Bank Machines: This parameter defines the number of bank machines. A given bank machine manages a single DRAM bank at any given time. 8

ORDERING: Normal mode allows the memory controller to reorder commands to the memory to obtain the highest possible efficiency. Strict mode forces the controller to execute commands in the exact order received. Strict

Memory Details: 1Gb, x16, row:13, col:10, bank:3, data bits per strobe:8, with data mask, single rank, 1.35V,1.5V

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Figure 2 - MIG Settings

Of particular importance is the clock period and the PHY to Controller clock ratio. The DDR3 chip you have is rated to operate between 300MHz – 400 MHz (DDR3-600 to DDR3-800), however, the memory controller on the Spartan 7 tops out at 333 MHz, which is what we will use. Since it is unlikely that the rest of your logic will run this quickly, the MIG provides a synchronous clock divider of either 4:1 or 2:1 and the AXI bus will run at this speed. For now, we will use 4:1 (giving us an AXI bus clock of 83.33MHz).

Click Next, and customize the settings as follows for the next page:

Memory Interface Generator

VIVADO
ML Editions

Pin Compatible FPGAs
Memory Selection
Controller Options
AXI Parameter
Memory Options
FPGA Options
Extended FPGA Options
IO Planning Options
Pin Selection
System Signals Selection
Summary
Simulation Options
PCB Information
Design Notes

Axi Parameter Options C0 - DDR3 SDRAM

Data Width
AXI DATA WIDTH: Data width of AXI read & write channels. The data width is less than or equal to user interface data width with the possible values 32, 64, 128, 256 & 512.

Arbitration Scheme
Select the arbitration scheme between the read and write address channels

Narrow Burst Support
Enables logic to support narrow bursts on the AXI4 slave interface. Can be set to zero if no masters in the system issue narrow bursts and all the data widths are equal. (1-Enable, 0-Disable)

Address Width
AXI4 address width of read and write address channels.

ID Width
AXI4 ID width for read and write channels. AXI4 ID is used as the identification tag for write or read address group of signals

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Figure 3 - MIG Settings Part 2

Note that by dividing down the AXI bus speed, the transfer rate to the memory controller will also decrease. However, we can recover some of the bandwidth by increasing the width of the AXI read and write channel signals. Since the MicroBlaze processor will only use a 32-bit data channel, we will keep this as 32 bits, however, if you are interfacing with the AXI bus directly in your SystemVerilog design, you may wish to use a wider channel. Click Next and move onto the following screen. Note that this next page overflows the UI, so you may need to scroll down to see all the options. Configure the design as follows:

Memory Options C0 - DDR3 SDRAM

Input Clock Period: Select the period for the PLL input clock (CLKIN). MIG determines the allowable input clock periods based on the Memory Clock Period entered above and the clocking guidelines listed in the User Guide. The generated design will use the selected Input Clock and Memory Clock Periods to generate the required PLL parameters. If the required input clock period is not available, the Memory Clock Period must be modified.

9750 ps (102.564 MHz)

☒ Select Additional Clocks (if required)

MIG can generate up to 5 additional clocks to be used in Fabric logic. This will be generated from the same MMCM which is used for generation of UI_CLK. The first clock (Clock 0) has a wider range of choices. All the values in the additional clocks drop downs are calculated considering the MMCM VCO frequency as **1501.50146 ps (666 MHz)** Mhz. For complete details on clocking of MIG, refer to MIG User Guide.

Clock	Period	D
Clock 0	5067 ps (197.33333 Mhz)	D = 3.375
Clock 1	NONE	D = 1
Clock 2	NONE	D = 1
Clock 3	NONE	D = 1
Clock 4	NONE	D = 1

Choose the Memory Options for the memory device. Memory Option selections are restricted to those supported by the controller. Consult the memory vendor data sheet for more information.

Read Burst Type and Length

The burst type determines the data ordering within a burst. Consult the memory datasheet for more information. Burst length 8 is the only supported value.

Sequential

Output Driver Impedance Control

Programmable impedance for the output buffer.

RZQ/7

RTT (nominal) - On Die Termination (ODT)

Select the nominal value of ODT for the DQ, DQS/DQS# and DM signals on the component or DIMM interface. This must be set to RZQ/6 ((40 ohms) for data rates at 1333 Mbps and above. In 2 slot DIMM configurations this value will be used for the unwritten slot during a write and will also be used for the unselected slot during a read. Use board level simulation to choose the optimum value.

RZQ/4

Controller Chip Select Pin

The Chip Select (CS#) pin can be tied low externally to save one pin in the address/command group when this selection is set to 'Disable'. Disable is only valid for single rank configurations.

Disable

Memory Address Mapping Selection

User Address

☐ ROW BANK COLUMN

☒ BANK ROW COLUMN

XILINX

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Figure 4 - MIG Settings 3

Of interest here is the input clock and the generation of additional clocks. Note that because 333.33 MHz (3000 ps clock) is not an even factor from 100 MHz, we must configure the MIG to 'expect' a 102.564 MHz clock instead of the actual 100 MHz oscillator. In practice, this means our design runs slightly slower than 333.33 MHz. Of note are the additional clocks, where we ask that the MIG additionally generate a 200 MHz (approximate) clock. As the MIG configures its own Clocking Wizard IP anyway, we can use this option to generate further clocks as necessary to reduce the total number of independent PLLs and MMCMs blocks used on the FPGA. The 200 MHz clock is used for the MIG's own calibration (training) process, you'll see that once we go to connect the overall Block Design.

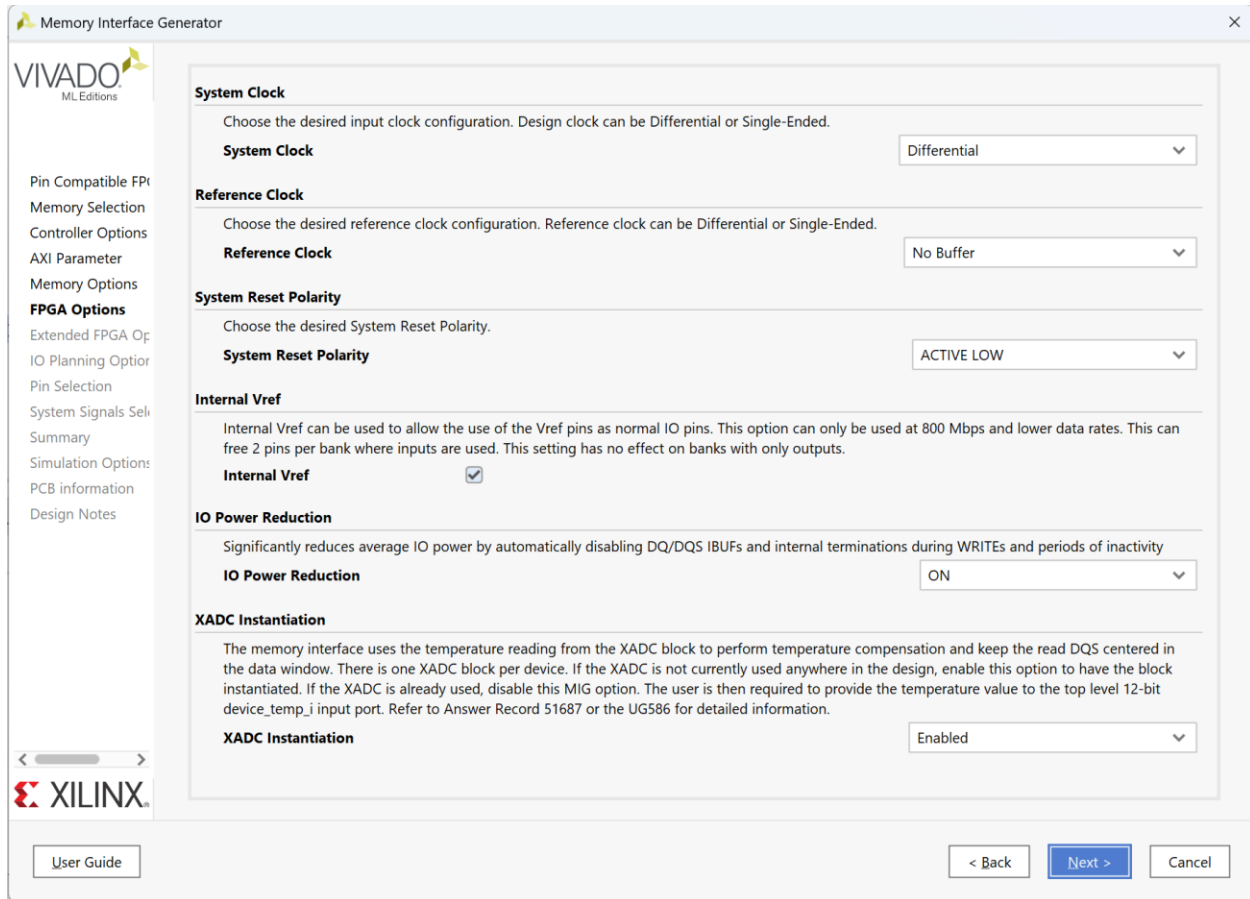


Figure 5 - MIG Settings 4

The next page has us configure some options related to the MIG as it is generated on the FPGA. Note that unlike in previous labs, we will be using the differential clock input. This allows for a more precise clock, and it allows for the onboard clock resources on the FPGA to lock faster. Notice also that the ‘reference clock’ is chosen to be “No Buffer”. This is for the 200 MHz clock, since we are internally generating this reference. The next screen just asks for the Termination Impedance, which we will keep as the default (50 Ohm), so click Next again.

We are prompted as to whether we want to start a new design or use an existing design. We would create a new design if the Urbana Board was still in the planning phase, and we wanted to best optimize the IO utilization of the Spartan 7 FPGA. However, since the Urbana Board is already designed and realized, we should select “Fixed Pin Out”.

The next screen prompts us to load a .XDC file to verify that the pin assignments for the design are compatible with the MIG. Go ahead and load the provided “urbana_ddr3.xdc” file and click Validate:

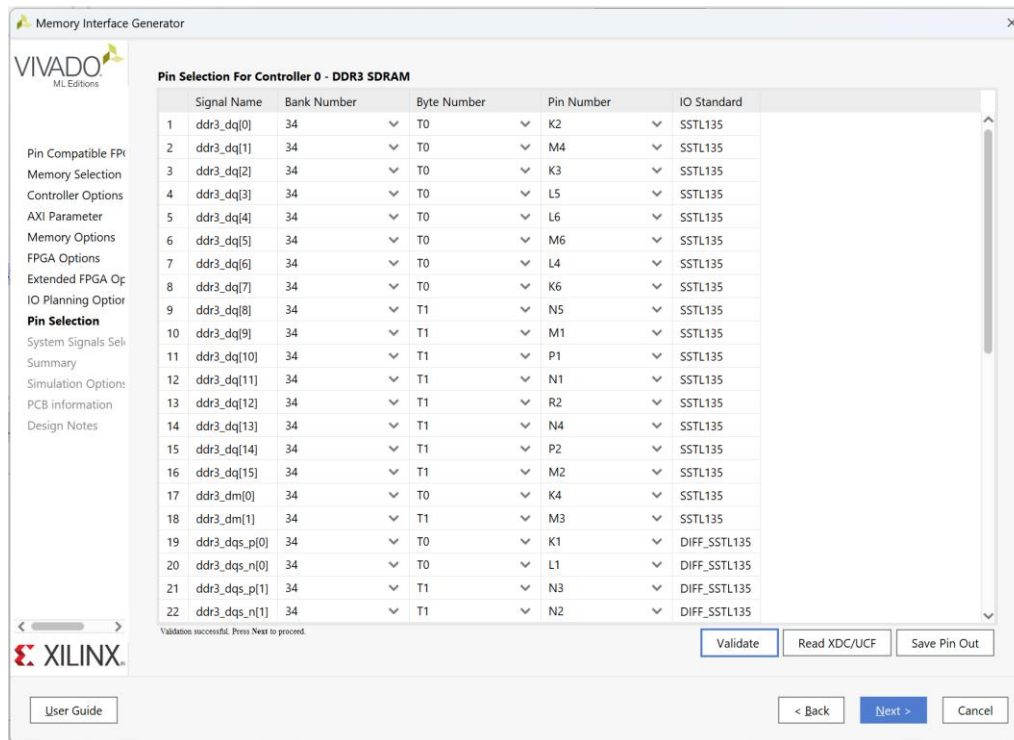


Figure 6 - XDC Validation

You should then be allowed to proceed, so click Next:

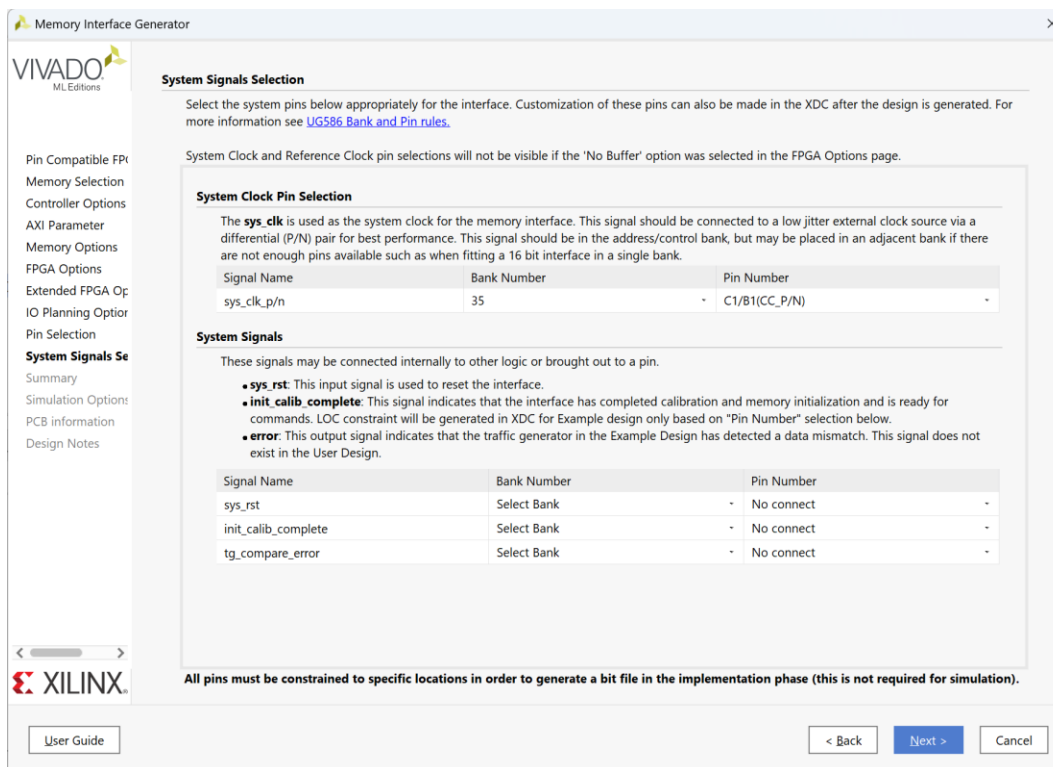


Figure 7 - Clock Selection

Now you are prompted to select the clock, which comes in on pins C1 and B1 (you may verify this with the schematic if you'd like). The other signals are optional, and therefore are not connected by default. You will then be prompted with a summary, and license agreement which you must click “Agree” and then another summary before you are prompted to click “Generate”.

Eventually you will be returned to the Block Design editor. Now you can click “Run Block Automation”. Instead of the Microcontroller preset, we will now use the Application processor preset, and configure as follows:

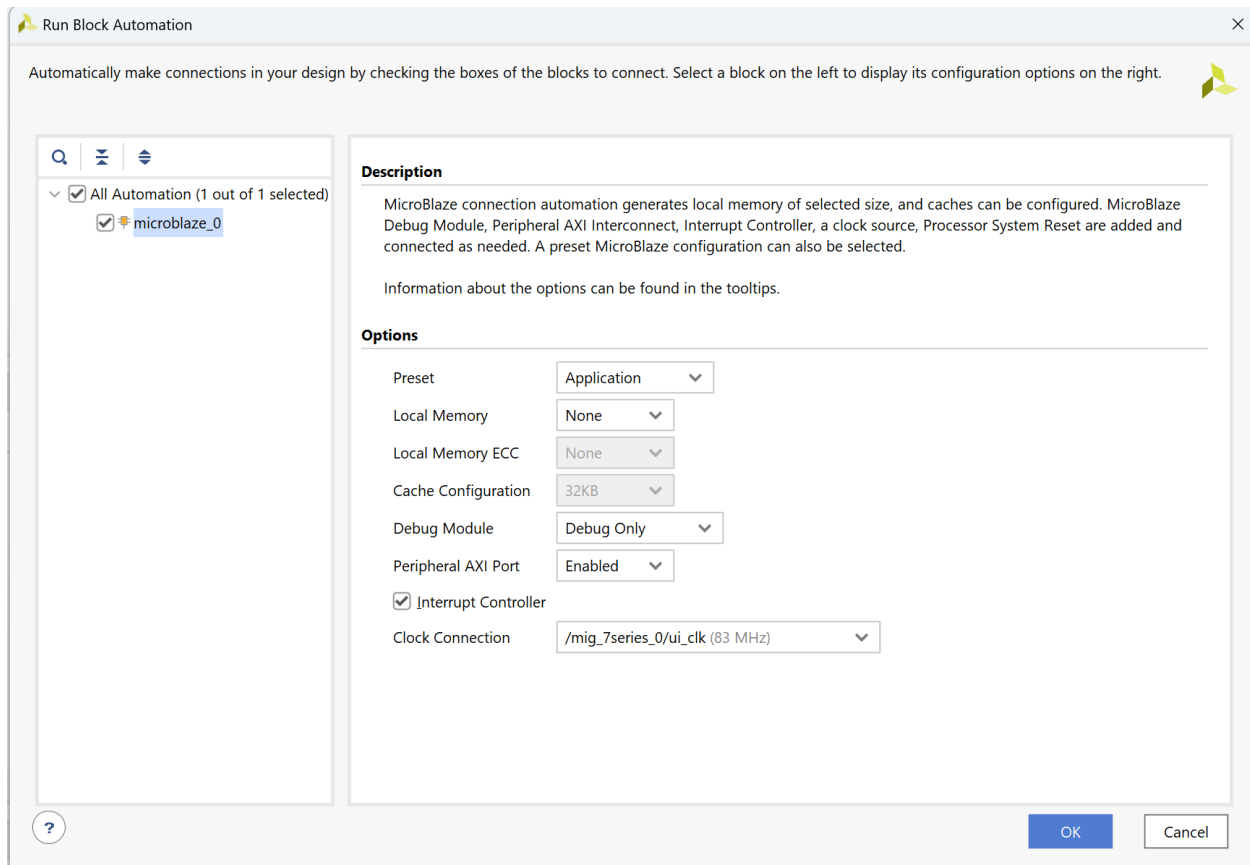


Figure 8 - Block Automation

Once you click OK, some connections will be made. Before proceeding further, double click on the MicroBlaze processor and deselect Enable Exceptions and Use Memory Management. This will simplify the CPU design by removing the exception handler and the MMU. The rest of the MicroBlaze settings may be left as their “Application” preset defaults. Obviously, if you are planning to run Linux on MicroBlaze, then you will need to keep the exception handler and the MMU.

Welcome to MicroBlaze Configuration Wizard



Usage Information

- Select a predefined configuration with *Select Configuration* below. Information about the selected configuration can be found in the tooltip. *Each predefined configuration completely changes the MicroBlaze parameters.*
- To modify the configuration, click on the *Next* button, click on the *Advanced* button at the top to directly access parameters in a tabbed interface, or click *OK* to accept the configuration and close the dialog.

Predefined Configurations

Select Configuration
Application Preset

Select Processor Implementation

☒ 32
☐ 64

General Settings

Select implementation optimization
PERFORMANCE

☒ Enable MicroBlaze Debug Module Interface
☒ Use Instruction and Data Caches
☐ Enable Exceptions
☐ Use Memory Management
☐ Enable Discrete Ports

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Figure 9 - MicroBlaze Settings

Now you should have the following Block Design or something similar:

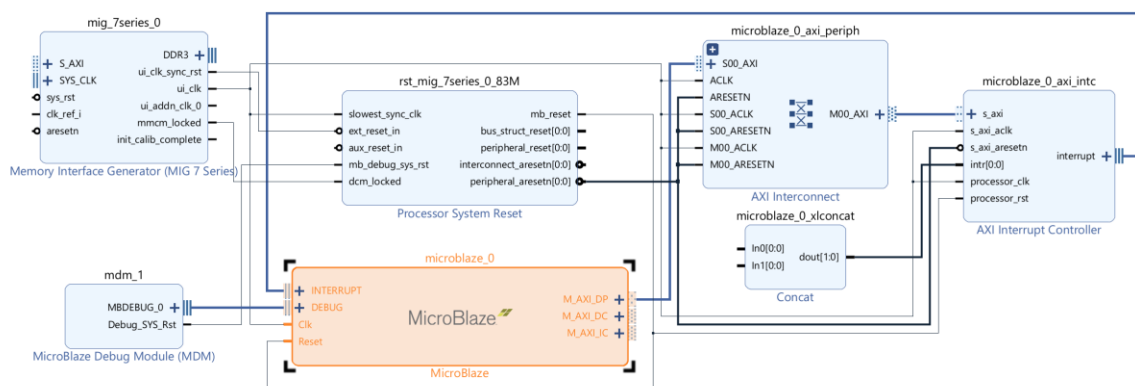


Figure 10 - After Connection Automation

There are still a few connections which need to be manually made. In addition, you should at the minimum instantiate two additional peripherals: the AXI timer, as well as the AXI Uartlite. Note that we have made the ddr3 port external, as well as the UART. Also, note that the 200 MHz clock we generated (ui_addn_clk_0) wires back to the MIG's reference clock (clk_ref_i). Finally, observe that the main AXI clock is now the 83.33MHz clock (ui_clk). Note that the external name for the clock is SYS_CLK, while the external port name for the DDR3 is ddr3 (lowercase) and the UART is uart (lowercase).

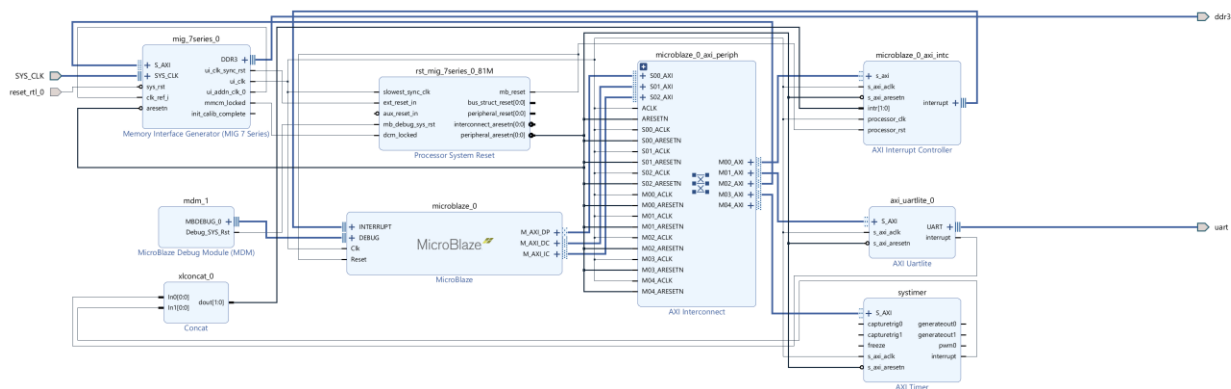


Figure 11 - Connected Block Design

Furthermore, the UART and the AXI Timer are configured as follows:

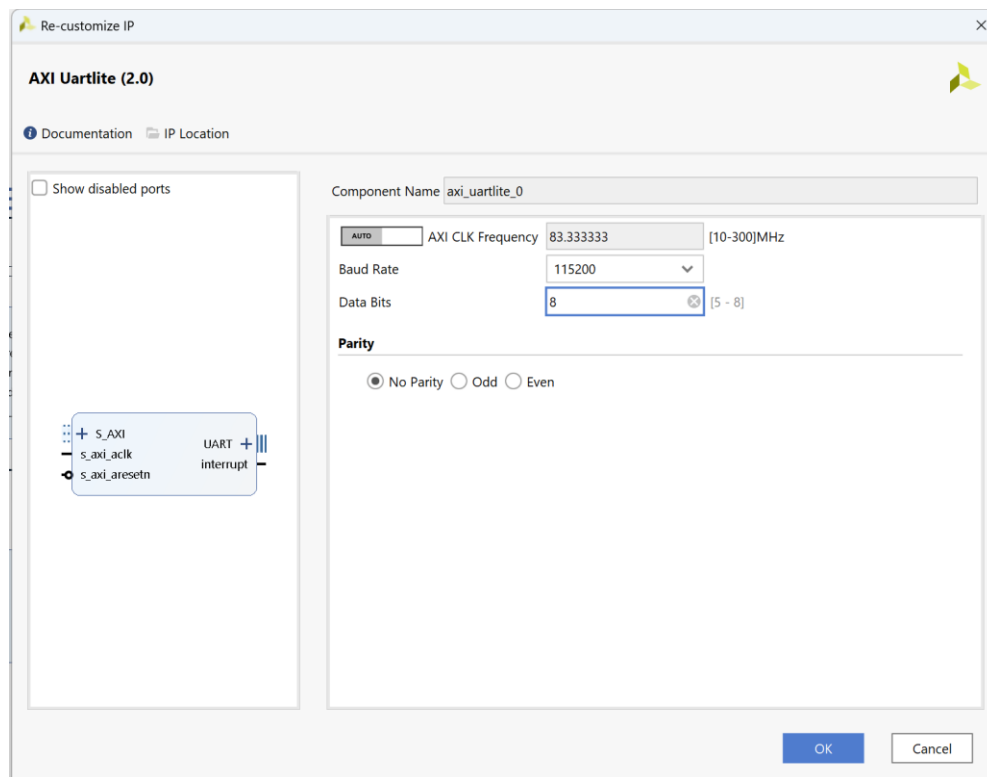


Figure 12 - UART Customization

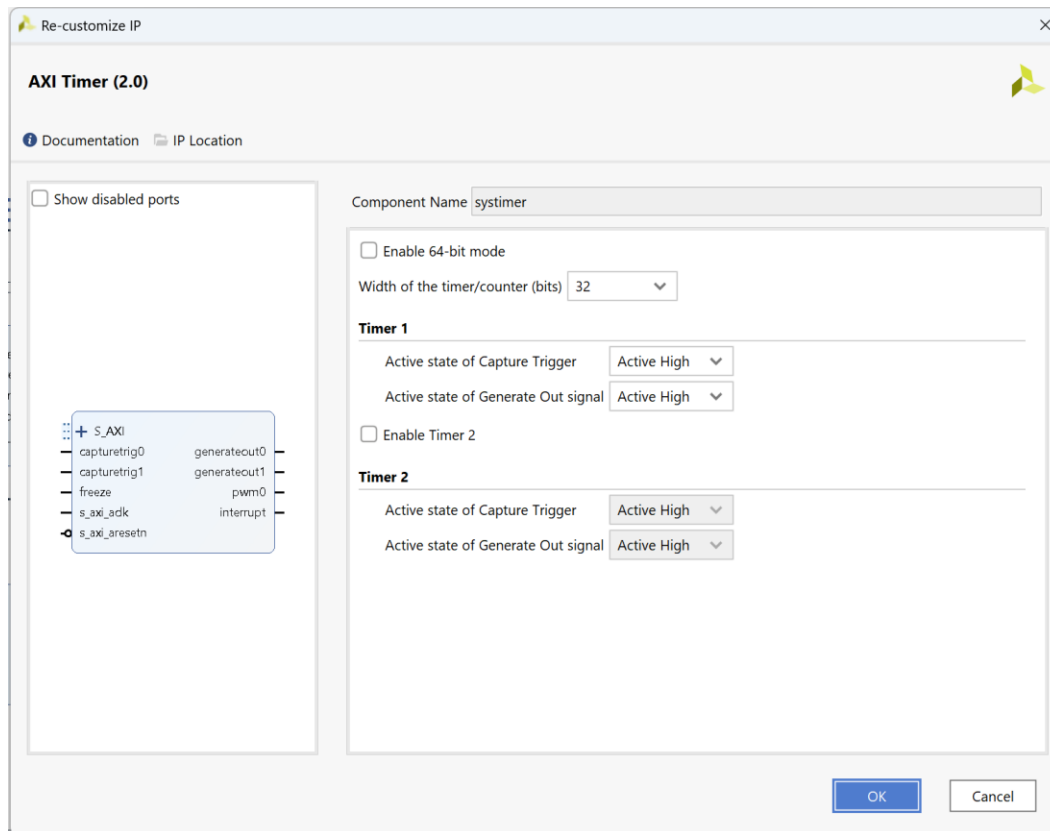


Figure 13 - Timer Customization

Finally, save and close the block design. Import the provided top level (mb_ddr3_top.sv) as well as the provided pin assignments (note that you already verified the contents, but they are still not imported into your project yet). Make sure to reconcile names between the top level and the Block Design as necessary (especially for the ports which have been made external). Once you can generate the bitstream without any errors, start Vitis as usual and make a new application project. Instead of selecting the “Hello World” template, instead select the “Dhrystone” template. This is a common CPU benchmark, which allows you to evaluate the MicroBlaze under different conditions (e.g., different amounts of cache, or performance vs. area optimizations).

Since you have gone through this process many times in this course, we will leave the student to refer to the previous documents for creating the Vitis project. Once the Dhrystone benchmark is properly compiling and running, the following should be displayed in the serial console, indicating that the MicroBlaze is properly executing code from DDR3.

```

Microseconds for one run through Dhrystone: 23.9265
Dhrystones per Second: 41794.7264
DMIPS/Sec: 23.7876
DMIPS/MHz: 0.2855

The Dhrystone App has run successfully

```

Figure 14 - Dhrystone Run